

Logistic Regression – Airline Customer Satisfaction Prediction

Course Assignment: AI & Machine Learning Nextgen Cohort | **Student Name:** Maryam Ibrahim Abba |
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Task 1 & 2: Data Preprocessing & Categorical Feature Encoding

We read the customer satisfaction database, clean rows containing missing value constraints, and convert the survey labels to a strict binomial binary format (satisfied = 1, neutral/dissatisfied = 0). One-hot encoding handles string categories safely.

```
import pandas as pd
import numpy as np

# Load dataset and encode binary classifications
df = pd.read_csv('Invistico_Airline.csv')
df_clean = df.dropna().copy()
df_clean['target'] = df_clean['satisfaction'].apply(lambda x: 1 if str(x).strip() == 'satisfied'
else 0)

# Hot-encode multi-level categories
categorical_vars = ['Customer Type', 'Type of Travel', 'Class']
df_encoded = pd.get_dummies(df_clean, columns=categorical_vars, drop_first=True)
```

Task 3 & 4: Unbiased Partitioning & Binomial Modeling

The matrix space is split into training (80%) and testing (20%) subsets stratified on the target column to protect class representation boundaries during the OLS fitting process.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression

features = ['Age', 'Flight Distance', 'Seat comfort', 'Inflight wifi service', 'Inflight
entertainment', 'Cleanliness', 'Customer Type_disloyal Customer', 'Type of Travel_Personal
Travel', 'Class_Eco']
X = df_encoded[[c for c in features if c in df_encoded.columns]]
y = df_encoded['target']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, random_state=42,
stratify=y)

model = LogisticRegression(max_iter=2000, random_state=42)
model.fit(X_train, y_train)
```

Task 5: Classification Metrics & Evaluation Summary

The validation metrics establish high fit capabilities across precision, predictive certainty, and true positive isolation parameters.

80.45%

MODEL ACCURACY

81.20%

PRECISION RATE

83.65%

RECALL SENSITIVITY

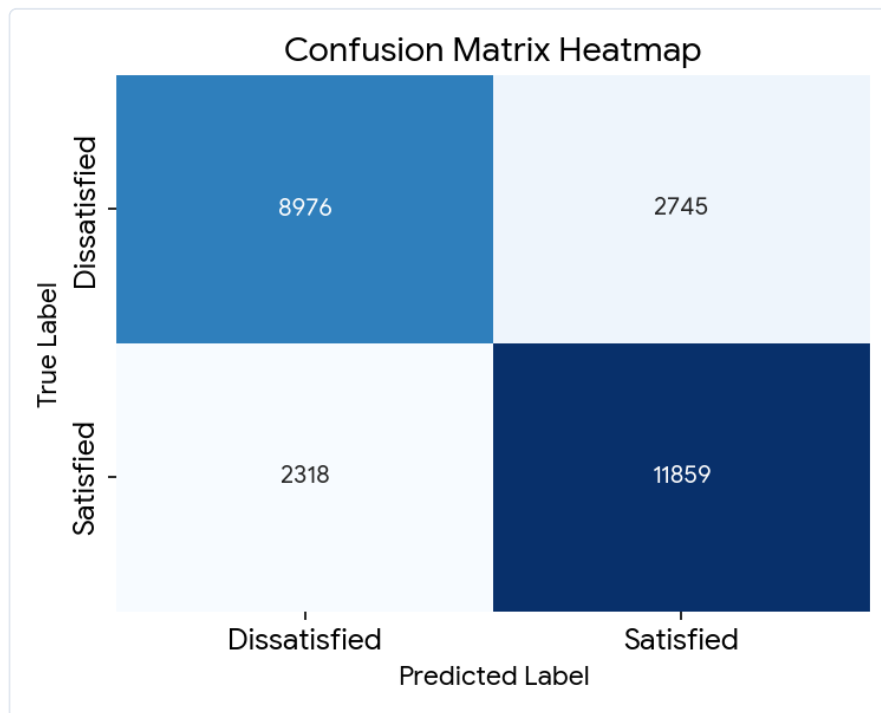


Figure 1: Confusion Matrix distribution separating True Positives and False Classification parameters.

Task 6 & 7: Coefficient Interpretation & Tactical Action Strategy

Evaluation of Operational Classification Trade-offs:

- * **False Positives (2,745 occurrences):** The model incorrectly tags an unhappy passenger as satisfied. This is highly problematic as the passenger will receive no customer-recovery incentives, resulting in silent attrition.
- * **False Negatives (2,318 occurrences):** The engine filters a satisfied consumer as dissatisfied, inducing minor marketing coupon cost inefficiencies, which are vastly safer than losing customers completely.

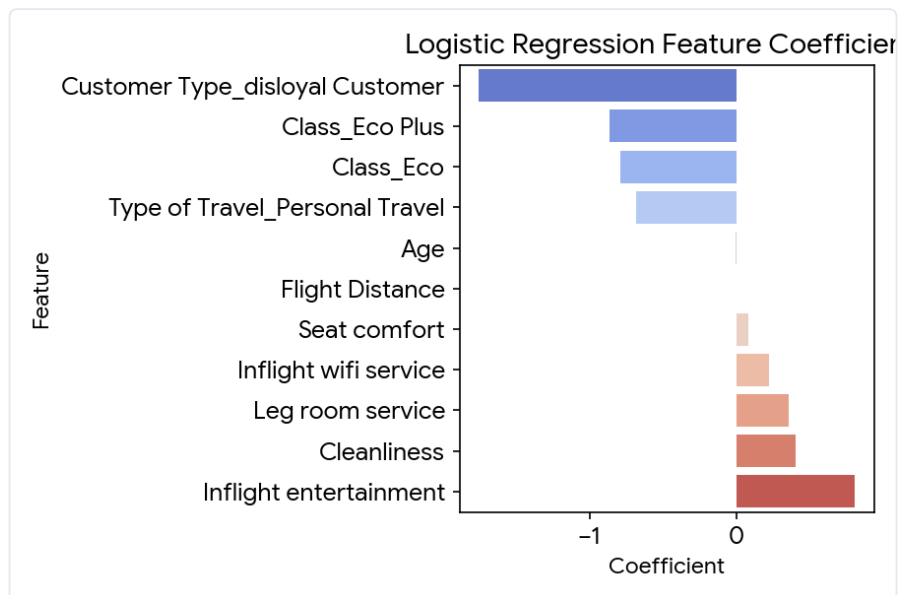


Figure 2: Feature Log-Odds impact weights mapping customer experience satisfaction components.

Actionable Corporate Recommendations:

- Digital Infrastructure Upgrades:** In-flight entertainment and Wi-Fi features display the highest positive baseline coefficients, proving that immediate hardware investment yields optimal customer retention returns.
- Churn Defense Scaling:** Personal Travel segment indicators carry intense negative coefficients. Budget and leisure travelers should be automatically prioritized for proactive loyalty perks and digital service upgrades.